HOTEL RECOMMENDATION SYSTEM

TEAM AND ROLES

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What is Hotel Recommendation System?

A hotel recommendation system is an AI-based tool that helps users find the best hotels based on their preferences. It uses various algorithms at backend to analyze data such as user ratings, reviews, location, price, and amenities to suggest hotels that meet the user's criteria. These systems can be integrated into travel websites, mobile apps, or used as standalone applications. They can also be personalized based on the user's past booking history and preferences.

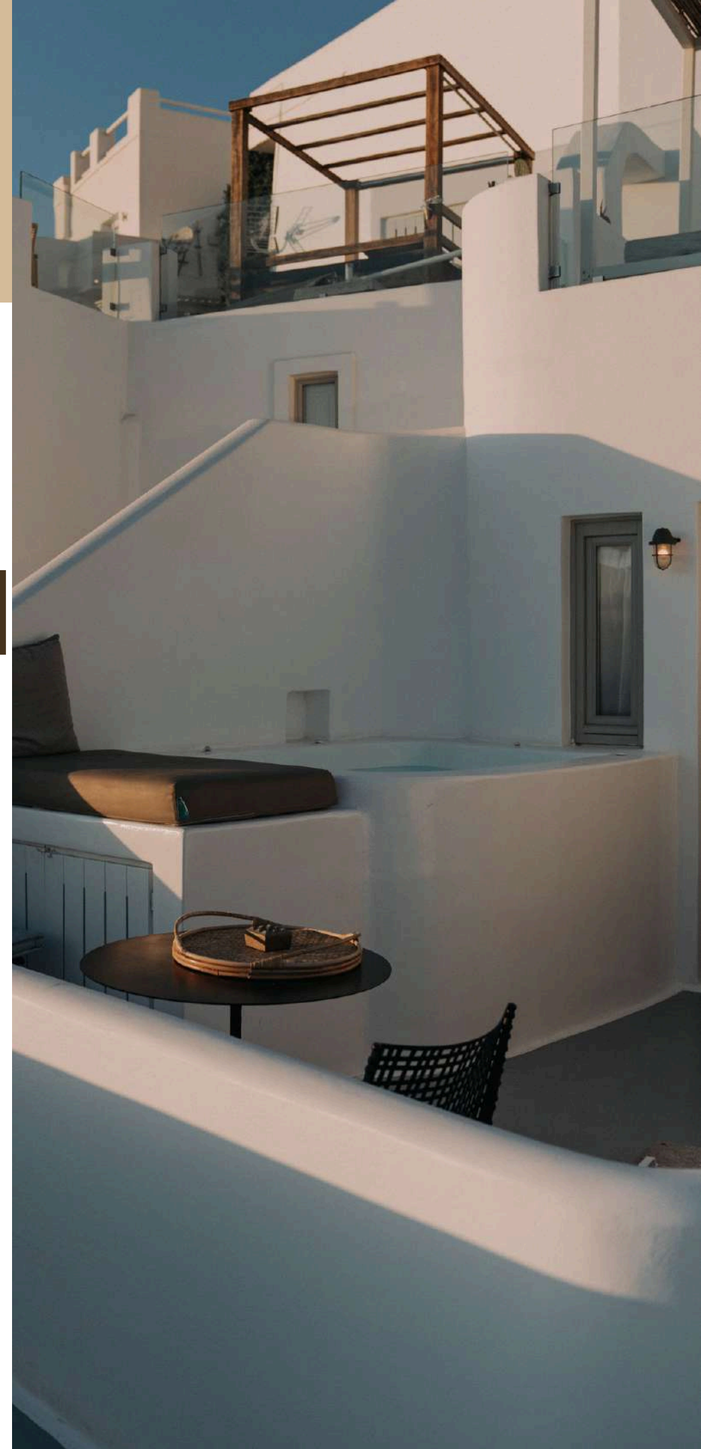
DATA COLLECTION

Data Collection Tool = KAGGLE

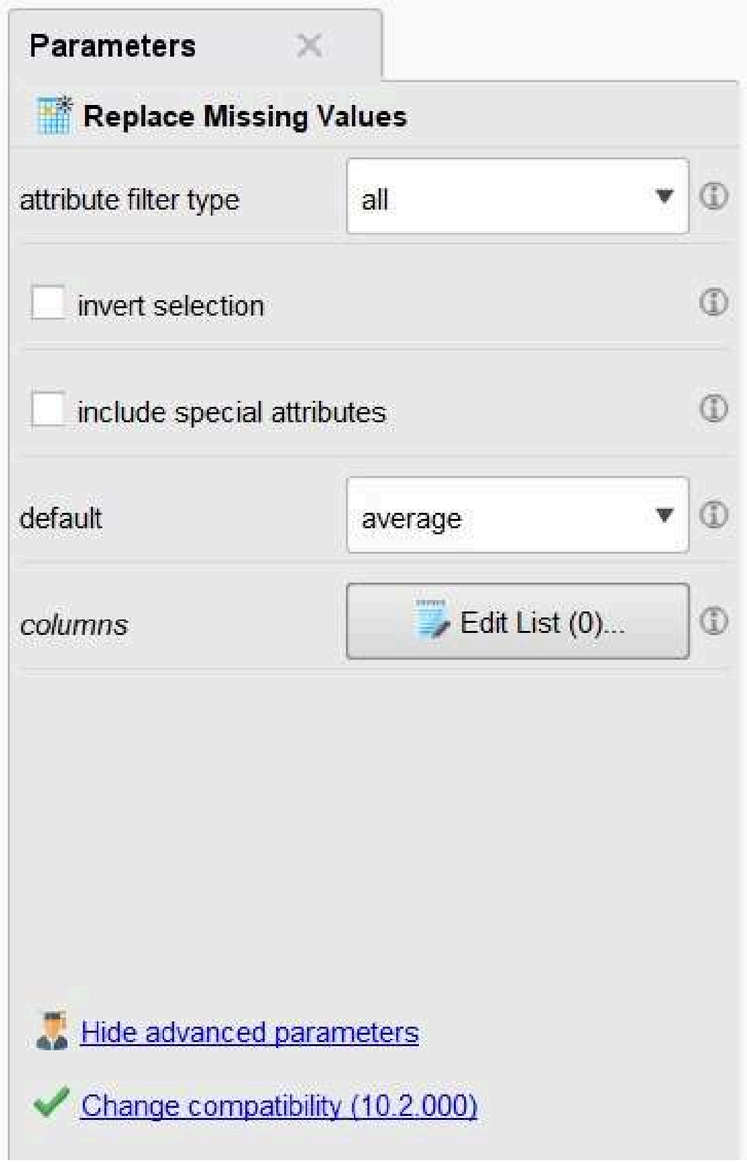
Dataset Name = Hotels In India Dataset [2023]

ATTRIBUTES NAME

- Hotel_id
- State
- City
- Ratings
- Alcohol Availability
- Hotel Name
- Total Rooms
- Category
- Pricing
- Meals



Data Preprocessing



Replace Missing Value

This Operator replaces missing values in Examples of selected Attributes by a specified replacement.

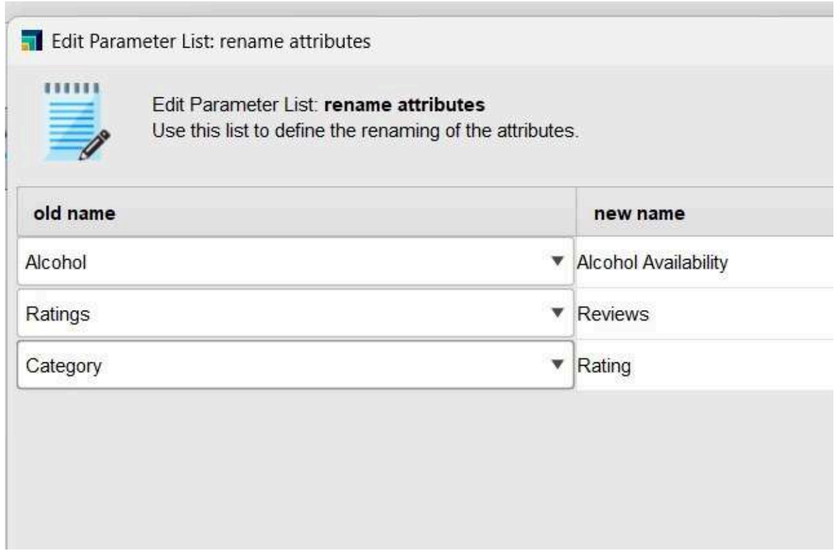
Here we have replaced the missing values with average values

Data Preprocessing

Rename

This operator can be used to rename one or more attributes of an ExampleSet.

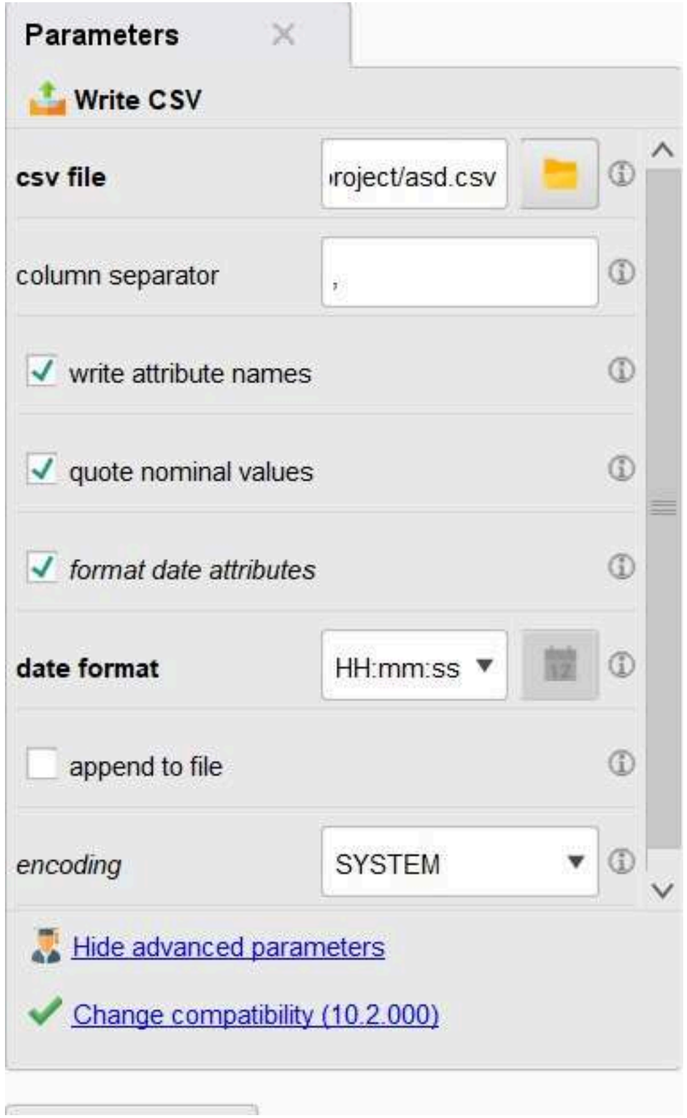
Here we have changed the Alcohol to Alcohol Availability
Rating to Reviews and Category to Rating



Write CSV

This operator is used to write CSV files(Comma-Separated Values).

This operator we have used to store the pre-processed csv file locally on our system

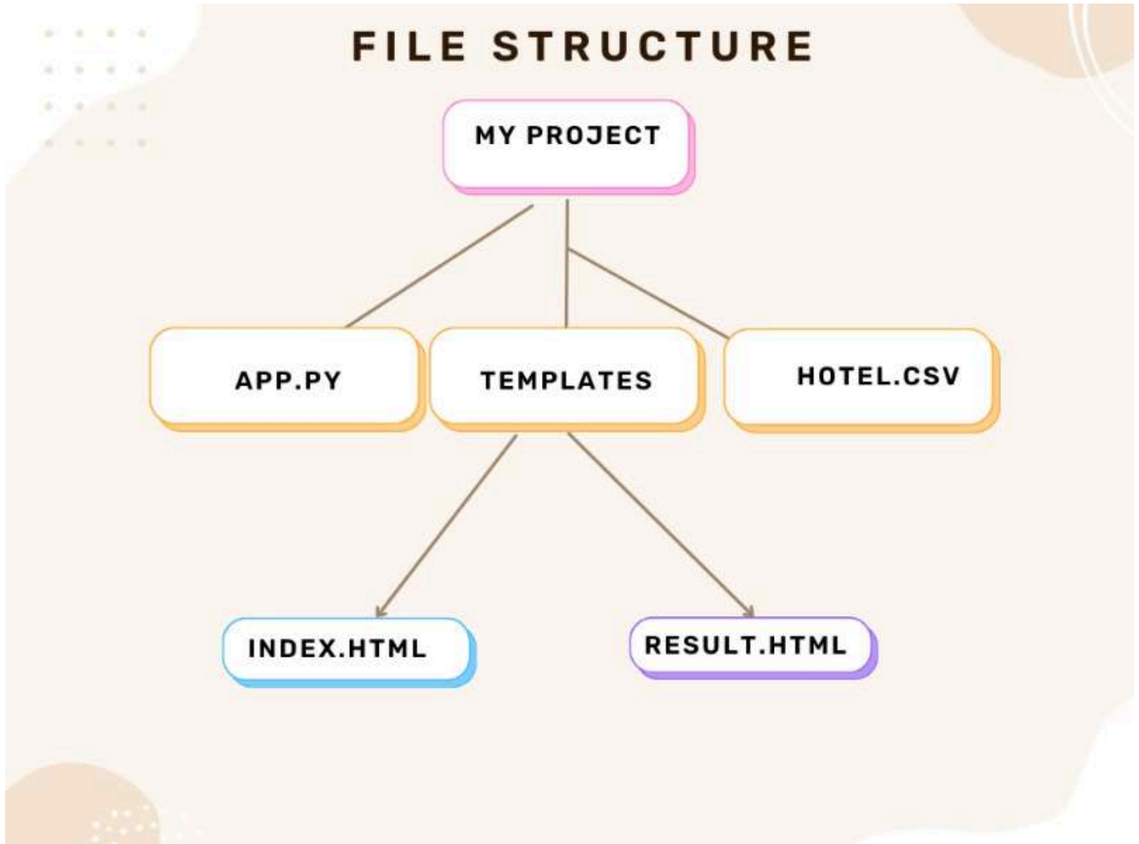


Data Preprocessing

This Code here is used to clean the data by removing special characters for it

```
Sub ReplaceSpecial()
    Dim cel As Range
    Dim strVal As String
    Dim i As Long
    Application.ScreenUpdating = False
    For Each cel In Selection
        strVal = cel.Value
        For i = 1 To Len(strVal)
            Select Case Asc(Mid(strVal, i, 1))
                Case 32, 48 To 57, 65 To 90, 97 To 122
                    ' Leave ordinary characters alone
                Case Else
                    Mid(strVal, i, 1) = " "
            End Select
        Next i
        cel.Value = strVal
    Next cel
    Application.ScreenUpdating = True
End Sub
```

PROJECT STRUCTURE



App.js

app.py

C: > Users > DELLS > hotel_recommend > app.py > recommend_hotels

```python
from flask import Flask, render_template, request
import pandas as pd
from sklearn.model_selection import train_test_split

app = Flask(__name__)

# Load the dataset and prepare it
df_hd = pd.read_csv("hotels.csv")
hotel_details = df_hd.iloc[:, [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]]

# Assume hotel_details is already prepared
X = hotel_details[['hotel_id', 'state', 'city', 'ratings', 'alcohol_availability', 'hotel_name', 'total_rooms', 'category', 'pricing', 'meals']]
y = hotel_details['hotel_name']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.26, random_state=42)

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/recommend_hotels', methods=['POST'])
def recommend_hotels():
    # Prompt user for input
    user_input = {
        'state': request.form['state'].lower(),
        'city': request.form['city'].lower(),
        'ratings': float(request.form['ratings']),
        'min_pricing': float(request.form['min_pricing']),
        'max_pricing': float(request.form['max_pricing'])
    }

    meals_option = request.form['meals_option']

    # Convert state and city to lowercase for case-insensitive comparison
    X_test['state'] = X_test['state'].str.lower()
    X_test['city'] = X_test['city'].str.lower()
```

```
36
37      # Apply filtering conditions
38      state_filter = X_test['state'] == user_input['state']
39      city_filter = X_test['city'] == user_input['city']
40      ratings_filter = X_test['ratings'] >= user_input['ratings']
41      pricing_filter = (X_test['pricing'] >= user_input['min_pricing']) & (X_test['pricing'] <= user_input['max_pricing'])
42
43      # Filtering by Meals Option
44      if meals_option == '1':
45          meals_filter = True  # No filtering based on meals
46      elif meals_option == '2':
47          meals_filter = X_test['meals'].str.contains('breakfast', case=False)
48      elif meals_option == '3':
49          meals_filter = X_test['meals'].str.contains('lunch', case=False)
50      elif meals_option == '4':
51          meals_filter = X_test['meals'].str.contains('dinner', case=False)
52      else:
53          print("Invalid option selected. Defaulting to 'All meals'.")
54          meals_filter = True
55
56      # Apply all filtering conditions
57      filtered_data = X_test[state_filter & city_filter & ratings_filter & pricing_filter & meals_filter]
58
59      if not filtered_data.empty:
60          recommended_hotels = filtered_data['hotel_id'].tolist()  # Get recommended hotel IDs
61          recommended_hotels_names = [y_test.loc[X_test[X_test['hotel_id'] == hotel_id].index[0]] for hotel_id in recommended_hotels]  # Get hotel names
62          return render_template('result.html', recommended_hotels=recommended_hotels_names)
63      else:
64          return "No hotels match the specified criteria."
65
66  if __name__ == '__main__':
67      app.run(debug=True)
```

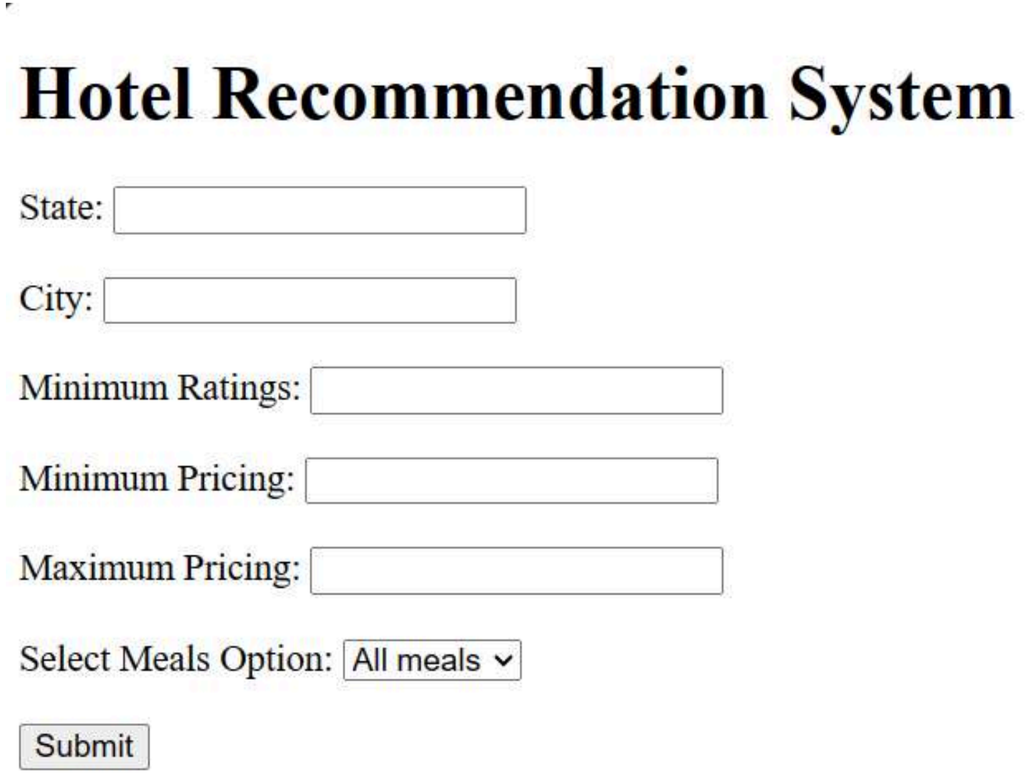
INDEX.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Hotel Recommendation</title>
<link rel="stylesheet" href="{{ url_for('static', filename='styles.css') }}">
</head>
<body>
    <h1>Hotel Recommendation</h1>
    <form action="/recommend_hotels" method="post">
        <label for="state">State:</label>
        <input type="text" id="state" name="state" required><br><br>

        <label for="city">City:</label>
        <input type="text" id="city" name="city" required><br><br>

        <label for="ratings">Minimum Ratings:</label>
        <input type="number" id="ratings" name="ratings" step="0.01" required><br><br>

        <label for="min_pricing">Minimum Pricing:</label>
        <input type="number" id="min_pricing" name="min_pricing" step="0.01" required><br><br>

        <label for="max_pricing">Maximum Pricing:</label>
        <input type="number" id="max_pricing" name="max_pricing" step="0.01" required><br><br>

        <label for="meals_option">Select Meals Option:</label>
        <select id="meals_option" name="meals_option" required>
            <option value="all_meals">All meals</option>
            <option value="breakfast">Breakfast</option>
            <option value="lunch">Lunch</option>
            <option value="dinner">Dinner</option>
        </select><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

RESULT.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Hotel Recommendation Result</title>
</head>
<body>
    <h1>Recommended Hotels</h1>
    <ul>
        {% for hotel_name in recommended_hotels %}
            <li>{{ hotel_name }}</li>
        {% endfor %}
    </ul>
</body>
</html>
```

OUTPUT


Hotel Recommendation System
State:
City:
Minimum Ratings:
Minimum Pricing:
Maximum Pricing:
Select Meals Option:
All meals
Submit

Hotel Recommendation System

State: Gujarat

City: Ahmedabad

Minimum Ratings: 2

Minimum Pricing: 11000

Maximum Pricing: 1500000

Select Meals Option: All meals

Submit

Recommended Hotels

- HOTEL NALANDA
- Gulmohar Greens Golf Country Club Limited
- BINORI HOTEL PVT LTD
- AJU IMPERIAL HOTEL A UNIT OF AJU RYOKAN NCR PVT LTD

Thank you!